

This assignment will not be graded and is only for practice.

Level 1

1) Consider two vectors $a = (2, 0, 3, 0, 8)$, $b = (3, 2, -2, 4, 0)$ in $\mathbb{R}^5$. Choose the set of correct options. **1 point**

- ☐ a and b are orthogonal.
- ☐ a and b are not orthogonal.
- ☐ $(a - b) \cdot a = 0$.
- ☐ $(a - b) \cdot a = 77$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

a and b are orthogonal.
 $(a - b) \cdot a = 77$.

2) Choose the set of correct statements.

1 point

- ☐ In an orthogonal set, the norms of all the vectors are equal.
- ☐ In an orthogonal set, the vectors are linearly independent.
- ☐ In an orthogonal set, the vectors are linearly dependent.

☐ If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in $\mathbb{R}^n$, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.

☐ If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in $\mathbb{R}^n$, then there are no solutions to the system $AX = b$, where X, b are $n \times 1$ vectors.

☐ The determinant of a square matrix formed by a set of orthogonal vectors in $\mathbb{R}^n$ is zero.

☐ A set of n vectors can never form an orthogonal basis in $\mathbb{R}^{n-1}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

In an orthogonal set, the vectors are linearly independent.

If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in $\mathbb{R}^n$, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.

A set of n vectors can never form an orthogonal basis in $\mathbb{R}^{n-1}$.

3) Which of the following is an orthogonal basis of the given vector spaces with respect to the standard inner product (dot product)?

1 point

- ☐ $\{(1, 0), (0, 1)\}$ is an orthogonal basis of $\mathbb{R}^2$.
- ☐ $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of $\mathbb{R}^3$.
- ☐ $\{(3, 4), (4, -3), (2, -3)\}$ is an orthogonal basis of $\mathbb{R}^2$.
- ☐ $\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$ is an orthogonal basis of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0), (0, 1)\}$ is an orthogonal basis of $\mathbb{R}^2$.

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of $\mathbb{R}^3$.

Level 2

4) Find a vector in $\mathbb{R}^4$ that is orthogonal to the subspace spanned by $(1, 1, 0, 0)$ and $(0, 1, 1, 0)$ with respect to the dot product as the inner product. **1 point**

☐ $(1, -1, 1, 0)$

☐ $(2, 3, 4, 5)$

☐ $(1, -1, 1, 1)$

☐ $(1, 1, 1, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, -1, 1, 0)$

$(1, -1, 1, 1)$

5) Which of the following are orthogonal to the vector $(1, 2)$ in $\mathbb{R}^2$ with respect to the inner product $\langle v, w \rangle = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$? **1 point**

☐ $(-7, 1)$

☐ $(6, -1)$

☐ $(7, 1)$

☐ $(-9, 1)$

☐ $(14, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, 1)$

$(14, 2)$

6)

Consider the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9.$$

Number of solutions of the above system is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

7) Let A be the coefficient matrix of the system of linear equations:

1 point

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9.$$

Which one of the following is true about the matrix AA^T ?

- ☐ A scalar matrix.
- ☐ The identity matrix.
- ☐ A diagonal matrix.
- ☐ A lower triangular matrix.
- ☐ An upper triangular matrix.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A diagonal matrix.

A lower triangular matrix.

An upper triangular matrix.

Your score is: 0/7